Project: Worked examples
Title: O-ring and seal gland
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1. Scope and method

Squeeze, stretch, gland fill and extrusion gap of a radial piston, radial rod or face O-ring seal, from the ring and groove dimensions with their tolerances, at the cold and shrunk and at the hot and swollen end of the temperature and swell range. The ring's cross-section shrinks with stretch by a relation you choose (none, constant-volume rubber, the quadratic cited from Parker ORD 5700 section 3.5, or linear in stretch). The worst case and the capability indices run through the tolerance-stack engine of M1, and the seeded Monte Carlo of the engine runs the exact non-linear model and reports the fraction outside each limit. A face seal is entered with the pressure on its outside or its inside. Measured groove distributions (from the scan import) can replace the toleranced dimensions. Ring sizes, the squeeze and fill bands and the extrusion limit are your own records: no O-ring size table, gland table or extrusion chart is shipped.

WHAT IT CALCULATES	RELATION
Ring stretch	$X = (g \cdot D_{ref})/(f \cdot d) - 1$, D_{ref} the diameter the ring ID sits on (groove root for a piston or face seal, rod for a rod seal); $f = (1 + \alpha(T - 20\ °C))(1 + \text{swell})^{(1/3)}$ the ring and $g = 1 + \alpha_m(T - 20\ °C)$ the groove metal
Ring cross-section	$W_{eff} = f \cdot W \cdot (1 - r(X))/\sqrt{(1 - c)}$, r(X) the chosen relation (none; $1 - 1/\sqrt{(1+X)}$; $Y = 0.56 + 0.59X - 0.0046X^2$ per cent as cited, used for about 3 to 25 % stretch only; $v \cdot X$) and c the compression of the ring outside diameter where it is held (rod seal in the housing groove; internal-pressure face seal on the groove outer wall), $c = 1 - g \cdot D_g/(f \cdot (d + 2W))$
Depth and gap	piston $h = (D_s - D_g)/2$, $\text{gap} = D_s - D_c$; rod $h = (D_g - D_s)/2$, $\text{gap} = D_c - D_s$; face h and gap entered directly (all times g)
Ring squeeze	$q = 1 - g \cdot h/W_{eff}$, minimum at the cold end from the largest depth over the smallest W_{eff}, maximum at the hot end
Gland fill	ring area over the groove area $g \cdot b \cdot g \cdot h - 2(1 - \pi/4)(g \cdot r_g)^2$; maximum at the hot end on a volume basis, $(\pi/4) \cdot (f \cdot W)^2 \cdot (f/g)$ with $f^3 = (1 + \alpha\Delta T)^3(1 + s)$ (an outside diameter held in the groove: $(\pi/4) \cdot (f \cdot W)^2/(1 - c)$), minimum at the cold end $(\pi/4) \cdot W_{eff}^2$
Worst case	each range from the engine's worst-case stack of the dimension ($\Sigma\lvert\text{coefficient}\rvert \cdot$half-band), combined at the corner that is worst for the quantity
Monte Carlo	the engine's seeded Monte Carlo on the exact criterion, each dimension from its own distribution; the fraction below the lower and above the upper limit. The capability index C_{pk} uses the criterion linearised about the mean dimensions (secant slope across each band)

WHAT IT CALCULATES	RELATION
Extrusion gap	the largest gap over the installed, cold and hot temperatures (the sealing-surface part and the groove metal each with its own expansion) plus the bore dilation under pressure, against the permitted gap from your record (interpolated in hardness and pressure) or the value entered; with neither the check fails (not verified)

Basis: Cross-section reduction with stretch (relation quoted second-hand; not confirmed)[1](Section 3.5); O-ring installation guidance (stretch rule quoted second-hand; not confirmed)[2]; O-rings: inside diameters, cross-sections, tolerances and size codes[3]; O-rings: housing dimensions for general applications[4].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Gland arrangement	`Piston (radial)`		spec: Cylinder datasheet LC-63: ring on the piston, sealing on the bore	any
duty	Static or dynamic duty (selects the squeeze band)	`Static`		spec: Cylinder datasheet LC-63: locking duty, no stroke under pressure	any
$face_p$	Face seal only: which side the pressure acts on (external: the ring ID seats on the groove inner wall and is stretched; internal: the ring OD seats on the groove outer wall and is compressed)	`External (ring ID seated on the groove)`		default	any
d_r	Ring inside diameter d (mean of the supplier's tolerance)	56	mm	reference: record: oring.ring, illustrative ring size invented for this example, to be replaced by the supplier's catalogue size	3 … 500 mm
$d_{r,tol}$	Ring inside diameter tolerance, plus or minus	0.45	mm	reference: record: oring.ring, illustrative tolerance invented for this example	0 … 2 mm
W	Ring cross-section diameter W	3.5	mm	reference: record: oring.ring, illustrative cross-section invented for this example	1 … 10 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
W_{tol}	Ring cross-section tolerance, plus or minus	0.1	mm	reference: record: oring.ring, illustrative tolerance invented for this example	0 … 0.4 mm
D_s	Sealing-surface diameter: the bore of a piston seal, the rod of a rod seal (not used for a face seal)	63	mm	drawing: LC-63-001 rev B	5 … 1000 mm
$D_{s,u}$	Sealing-surface diameter, upper deviation	0.046	mm	drawing: LC-63-001 rev B	-0.5 … 0.5 mm
$D_{s,l}$	Sealing-surface diameter, lower deviation	0	mm	drawing: LC-63-001 rev B	-0.5 … 0.5 mm
D_g	Groove root diameter: on the piston, in the housing of a rod seal, or for a face seal the diameter the ring seats on (the groove inner wall for external pressure, the groove outer wall for internal pressure)	57.5	mm	drawing: LC-63-002 rev B	5 … 1000 mm
$D_{g,u}$	Groove root diameter, upper deviation	0	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
$D_{g,l}$	Groove root diameter, lower deviation	-0.074	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
D_c	Mating part that sets the extrusion gap: the piston outside diameter, or the housing bore round the rod (not used for a face seal)	63	mm	drawing: LC-63-002 rev B	5 … 1000 mm
$D_{c,u}$	Mating part diameter, upper deviation	-0.03	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
$D_{c,l}$	Mating part diameter, lower deviation	-0.06	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
h_f	Face-seal groove depth (face seal only; the ring is compressed over $h_f + e_f$)	2.7	mm	default	0.3 … 20 mm
$h_{f,u}$	Face-seal groove depth, upper deviation	0.05	mm	default	-0.5 … 0.5 mm
$h_{f,l}$	Face-seal groove depth, lower deviation	0	mm	default	-0.5 … 0.5 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
e_f	Face-seal gap between the flanges (face seal only): the extrusion gap, and it adds to the compression height h_f + e_f	0.05	mm	default	0 … 2 mm
$e_{f,u}$	Face-seal gap, upper deviation	0.05	mm	default	-0.5 … 0.5 mm
$e_{f,l}$	Face-seal gap, lower deviation	0	mm	default	-0.5 … 0.5 mm
b	Groove width b	4.7	mm	drawing: LC-63-002 rev B	1 … 20 mm
b_u	Groove width, upper deviation	0.1	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
b_l	Groove width, lower deviation	0	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
ecc	Gland eccentricity or run-out (one-sided offset; it deepens the gland by this much where the squeeze is least and shallows it where the squeeze is most; 0 if none)	0	mm	default	0 … 0.5 mm
r_g	Groove bottom corner radius	0.25	mm	drawing: LC-63-002 rev B	0 … 1 mm
T_{min}	Lowest operating temperature (the ring sits at 20 °C when installed)	-25	°C	spec: Hydraulic specification HS-63: oil temperature range −25 to +70 °C	-60 … 100 °C
T_{max}	Highest operating temperature	70	°C	spec: Hydraulic specification HS-63: oil temperature range −25 to +70 °C	-40 … 200 °C
α	Ring linear thermal expansion coefficient α (an indicative elastomer figure; take yours from the compound or the material library)	0.00015	1/K	assumption: Indicative elastomer figure; replace by the compound's value or a material from the library.	0.00005 … 0.0003 1/K
$α_m$	Groove metal linear thermal expansion coefficient $α_m$ (the part that carries the groove; one material record per sheet, so this one is entered from the metal's data sheet)	0.000012	1/K	assumption: Steel groove metal, indicative.	0.000005 … 0.00003 1/K

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
s_{min}	Smallest volume change of the ring in the fluid (negative for shrinkage; 0 if none)	0	%	assumption: No shrinkage assumed in the mineral oil.	-10 … 0 %
s_{max}	Largest volume swell of the ring in the fluid	4	%	assumption: Volume swell of the compound in the oil, assumed 4 %; replace by the compound's immersion data.	0 … 25 %
red	Cross-section reduction with stretch	Incompressible rubber (derived)		assumption: Constant-volume rubber relation, derived.	any
v	Poisson ratio of the ring, for the linear relation	0.49		default	0.3 … 0.5
$q_{lo,s}$	Static duty: lowest acceptable squeeze (from your handbook)	15	%	reference: record: oring.criteria, static squeeze band, to be taken from the owner's handbook or the seal maker	5 … 30 %
$q_{hi,s}$	Static duty: highest acceptable squeeze	30	%	reference: record: oring.criteria, static squeeze band, to be taken from the owner's handbook or the seal maker	10 … 45 %
$q_{lo,d}$	Dynamic duty: lowest acceptable squeeze	10	%	default	5 … 25 %
$q_{hi,d}$	Dynamic duty: highest acceptable squeeze	25	%	default	10 … 35 %
$fill_{lo}$	Gland fill: lowest acceptable	60	%	reference: record: oring.criteria, gland fill band (default 60 to 85 %)	40 … 80 %
$fill_{hi}$	Gland fill: highest acceptable	85	%	reference: record: oring.criteria, gland fill band (default 60 to 85 %)	60 … 95 %
x_{lo}	Installed stretch: lowest acceptable (negative allows a ring that is compressed on its seat)	0	%	reference: record: oring.criteria, installed stretch lower limit	-5 … 5 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
x_{hi}	Installed stretch: highest acceptable (the installation rule is up to 6 % on a piston groove); also applied at the cold end	6	%	reference: record: oring.criteria, installed stretch limit (up to 6 % on a piston groove, as quoted second-hand)	1 … 10 %
od_{hi}	Ring outside-diameter compression: highest acceptable, in a rod-seal groove or against the groove outer wall of an internal-pressure face seal (your record; the default is a placeholder equal to the stretch limit, not a handbook value)	6	%	default	1 … 10 %
H_A	Ring hardness, Shore A (for the extrusion record)	70		spec: Seal specification: compound 70 Shore A	40 … 95
p_{op}	Operating pressure (for the extrusion record, gauge)	160	bar g	spec: Hydraulic specification HS-63: working pressure 160 bar	0 … 700 bar g
dD_p	Diametral growth under the operating pressure of the bore (piston seal) or of the housing bore round the rod (rod seal), from your cylinder or housing calculation; added to the extrusion gap (0 if not known)	0	mm	default	0 … 0.5 mm
gap_{allow}	Largest permitted extrusion gap at this pressure and hardness, if you have it as one value: diametral (the clearance $D_s - D_c$) for a radial seal, axial for a face seal; a radial chart value is half of it	0.2	mm	reference: record: oring.extrusion, illustrative permitted gap invented for this example, to be replaced by the seal maker's value for 70 Shore A at 160 bar	0.01 … 1 mm
n_{mc}	Monte Carlo samples	20000		assumption: 20 000 samples.	1000 … 100000
seed	Monte Carlo seed	42		assumption: Fixed seed for a repeatable sheet.	0 … 4295000000

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_{sigma}	Standard deviations per half-band (normal tolerances and measured normals)	3		assumption: Normal tolerances taken as ±3σ.	1 ... 6
dist	Distribution of the toleranced dimensions	Normal		assumption: Normal distribution of the machined dimensions.	any

3. Assumptions

A1. The gland is a plain rectangular groove with a bottom corner radius; the ring is a circle of the stated cross-section. Edge breaks, groove wall angles and surface finish are not modelled. (Closed-form gland geometry; no gland table.)
A2. Squeeze is the compressed fraction of the effective cross-section W_{eff}, which is the nominal section after thermal expansion, swell and the cross-section reduction of the stretch. The reduction relation is chosen by the user; the quadratic of Parker ORD 5700 section 3.5 is implemented as quoted second-hand, used for about 3 to 25 % stretch only, and is not confirmed against the handbook. Where the ring outside diameter is held (a rod seal in its housing groove, an internal-pressure face seal), the compression c thickens the section to W/√(1 − c). (The relation is an input choice; the default is the constant-volume rubber relation, derived.)
A3. The cold end is the lowest temperature with the least swell and sets the minimum squeeze and fill; the hot end is the highest temperature with the most swell and sets the maximum. This assumes the ring expands more than the groove metal (α above α_m). (Bounding cases of a monotonic model.)
A4. The worst case combines each dimension at its extreme independently. Where one dimension appears twice (the groove root diameter sets both the depth and the stretch of a piston seal) the worst case is slightly conservative. (Independent ends of linear stacks.)
A5. The Monte Carlo samples each dimension from its own distribution about its mean and evaluates each criterion exactly (the non-linear model) on the engine's seeded sampler. It counts the fraction outside each limit at the cold or hot end; it does not model pressure, friction or compression set. The capability index C_{pk} uses the first-order (secant) linearisation. (Exact model on the engine's seeded sampler.)
A6. The gland fill at the hot end is on a volume basis: a ring held circumferentially by the groove takes the whole volume change of thermal expansion and swell in its section, $(1 + \alpha\Delta T)^3(1 + s)$, while the groove circumference grows with the groove metal. This is higher than scaling the section area isotropically. (Volume conservation of the held ring.)
A7. The extrusion gap is the largest diametral clearance of a radial seal (the axial gap of a face seal) over the installed, cold and hot temperatures, the sealing-surface part and the groove metal each with its own expansion coefficient, plus the diametral growth of the bore under pressure that the user enters (0 if none). The permitted gap is the user's record; the sheet never states a handbook value. With no permitted gap entered the extrusion check fails (not verified) while the operating pressure is above zero. (Extrusion limits are licensed data and are not shipped.)
A8. α (Ring linear thermal expansion coefficient α (an indicative elastomer figure; take yours from the compound or the material library)) is taken as 0.00015 1/K. (Indicative elastomer figure; replace by the compound's value or a material from the library.)
A9. α_m (Groove metal linear thermal expansion coefficient α_m (the part that carries the groove; one material record per sheet, so this one is entered from the metal's data sheet)) is taken as 0.000012 1/K. (Steel groove metal, indicative.)
A10. s_{min} (Smallest volume change of the ring in the fluid (negative for shrinkage; 0 if none)) is taken as 0 %. (No shrinkage assumed in the mineral oil.)
A11. s_{max} (Largest volume swell of the ring in the fluid) is taken as 4 %. (Volume swell of the compound in the oil, assumed 4 %; replace by the compound's immersion data.)
A12. red (Cross-section reduction with stretch) is taken as Incompressible rubber (derived).

(Constant-volume rubber relation, derived.)
A13. n_{mc} (Monte Carlo samples) is taken as 20000. (20 000 samples.)
A14. seed (Monte Carlo seed) is taken as 42. (Fixed seed for a repeatable sheet.)
A15. k_{sigma} (Standard deviations per half-band (normal tolerances and measured normals)) is taken as 3. (Normal tolerances taken as ±3σ.)
A16. dist (Distribution of the toleranced dimensions) is taken as Normal. (Normal distribution of the machined dimensions.)
A17. Static duty: the squeeze band for static seals from the oring.criteria record is used. (The bands are the user's record.)

4. Working

[42] v_W = mid ± half-band, σ (normal)
= 3.5 mm nominal → mid 3.5 ± 0.1 mm, σ 0.033333 mm
= 3.5 mm
ring cross-section W
[43] v_d = mid ± half-band, σ (normal)
= 56 mm nominal → mid 56 ± 0.45 mm, σ 0.15 mm
= 56 mm
ring inside diameter d
[44] v_{Ds} = mid ± half-band, σ (normal)
= 63 mm nominal → mid 63.023 ± 0.023 mm, σ 0.0076667 mm
= 63.023 mm
bore D_s
[45] v_{Dg} = mid ± half-band, σ (normal)
= 57.5 mm nominal → mid 57.463 ± 0.037 mm, σ 0.012333 mm
= 57.463 mm
piston groove root D_g
[46] v_{Dc} = mid ± half-band, σ (normal)
= 63 mm nominal → mid 62.955 ± 0.015 mm, σ 0.005 mm
= 62.955 mm
piston outside diameter D_c
[47] v_b = mid ± half-band, σ (normal)
= 4.7 mm nominal → mid 4.75 ± 0.05 mm, σ 0.016667 mm
= 4.75 mm
groove width b
[48] $f_c = (1 + \alpha\cdot(T_{min} - 20\ °C))\cdot(1 + s_{min})^{(1/3)}$
$= (1 + 0.00015\cdot(-25 - 20))\cdot(1 + 0)^{(1/3)}$
= 0.99325
Ring size factor, cold end
[49] $g_c = 1 + \alpha_m\cdot(T_{min} - 20\ °C)$
$= 1 + 0.000012\cdot(-25 - 20)$
= 0.99946
Groove metal size factor, cold end
[50] $f_h = (1 + \alpha\cdot(T_{max} - 20\ °C))\cdot(1 + s_{max})^{(1/3)}$
$= (1 + 0.00015\cdot(70 - 20))\cdot(1 + 0.04)^{(1/3)}$
= 1.0208
Ring size factor, hot end
[51] $g_h = 1 + \alpha_m\cdot(T_{max} - 20\ °C)$
$= 1 + 0.000012\cdot(70 - 20)$
= 1.0006
Groove metal size factor, hot end
[52] $stretch_{nom} = D_{ref}/d - 1$ at the mean dimensions
$= 57.463/56 - 1$
= 2.6125 %

Installed stretch of the ring ID on the seat, mean dimensions

$$[53]\ \text{stretch} = D_{ref,max}/d_{min} - 1$$
$$= 57.5/55.55 - 1$$
$$= 3.5104\ \%$$

Largest installed stretch

$$[54]\ \text{stretch}_{min} = D_{ref,min}/d_{max} - 1$$
$$= 57.426/56.45 - 1$$
$$= 1.729\ \%$$

Smallest installed stretch

$$[55]\ \text{red}_{nom} = r(X) \text{ at the mean installed stretch}$$
$$= r(2.6125\ \%)$$
$$= 1.2812\ \%$$

Cross-section reduction at the mean installed stretch

$$[56]\ u_{stretch} = \text{stretch} / x_{hi}$$
$$= 3.5104\ \% / 6\ \%$$
$$= 0.58506$$

Largest installed stretch against its limit

$$[57]\ X_c = g_c \cdot D_{ref,max}/(f_c \cdot d_{min}) - 1$$
$$= 0.99946 \cdot 57.5/(0.99325 \cdot 55.55) - 1$$
$$= 4.1575\ \%$$

Largest stretch at the cold end

$$[58]\ r_c = r(X_c)$$
$$= r(4.1575\ \%)$$
$$= 2.0161\ \%$$

Cross-section reduction at the cold end

$$[59]\ Wc_{lo} = f_c \cdot W_{min}$$
$$= 0.99325 \cdot 3.4$$
$$= 3.377\ \text{mm}$$

Smallest ring section at the cold end

$$[60]\ Weff_{lo} = Wc_{lo} \cdot (1 - r_c)$$
$$= 3.377 \cdot (1 - 0.020161)$$
$$= 3.309\ \text{mm}$$

Smallest effective ring section

$$[61]\ h_{hi,c} = g_c \cdot (\text{depth stack}) \text{ maximum}$$
$$= \text{mean } 2.7785\ \text{mm, worst-case maximum}$$
$$= 2.8085\ \text{mm}$$

Largest gland depth at the cold end

$$[62]\ \text{squeeze}_{min} = 1 - h_{hi,c}/Weff_{lo}$$
$$= 1 - 2.8085/3.309$$
$$= 15.125\ \%$$

Smallest squeeze

$$[63]\ A_{g,hi} = b_{max} \cdot h_{max} - 2(1 - \pi/4) \cdot (g_c \cdot r_g)^2$$
$$= 4.7974 \cdot 2.8085 - 0.026796$$
$$= 13.447\ \text{mm}^2$$

Largest groove area at the cold end

$$[64]\ \text{fill}_{min} = (\pi/4) \cdot Weff_{lo}^2 / A_{g,hi}$$
$$= (\pi/4) \cdot 3.309^2 / 13.447$$
$$= 63.953\ \%$$

Smallest gland fill

$$[65]\ X_h = g_h \cdot D_{ref,min}/(f_h \cdot d_{max}) - 1$$
$$= 1.0006 \cdot 57.426/(1.0208 \cdot 56.45) - 1$$
$$= -0.28\ \%$$

Smallest stretch at the hot end

[66] r_h = r(X_h)
= r(-0.28 %)
= -0.14029 %
Cross-section reduction at the hot end
[67] Wh_{hi} = $f_h \cdot W_{max}$
= 1.0208·3.6
= 3.6747 mm
Largest ring section at the hot end
[68] $Weff_{hi}$ = Wh_{hi}·(1 - r_h)
= 3.6747·(1 - -0.0014029)
= 3.6799 mm
Largest effective ring section
[69] $h_{lo,h}$ = g_h·(depth stack) minimum
= mean 2.7817 mm, worst-case minimum
= 2.7516 mm
Smallest gland depth at the hot end
[70] $squeeze_{max}$ = 1 - $h_{lo,h}/Weff_{hi}$
= 1 - 2.7516/3.6799
= 25.225 %
Largest squeeze
[71] $A_{g,lo}$ = $b_{min} \cdot h_{min}$ - 2(1 - π/4)·$(g_h \cdot r_g)^2$
= 4.7028·2.7516 - 0.026857
= 12.914 mm^2
Smallest groove area at the hot end
[72] $fill_{max}$ = (π/4)·$Wh_{hi}^2 \cdot (f_h/g_{ref,h})/A_{g,lo}$
= (π/4)·3.6747^2·(1.0208/1.0006)/12.914
= 83.783 %
Largest gland fill (unstretched section, volume basis: the held ring takes its whole volume change in its section)
[73] $squeeze_{nom}$ = 1 - h/W_{eff} at the mean dimensions
= 1 - 2.78/3.4552
= 19.541 %
Squeeze at the mean dimensions, installed
[74] $sq_{min,exact}$ = exact non-linear corner (check)
= corner chosen by the sensitivities
= 15.18 %
Check: smallest squeeze at the exact corner
[75] $sq_{max,exact}$ = exact non-linear corner (check)
= corner chosen by the sensitivities
= 25.176 %
Check: largest squeeze at the exact corner
[76] gap_{max} = largest gap over the installed, cold and hot temperatures (diametral clearance, with its own expansion) plus the bore dilation under pressure
= mean 0.068 mm, worst-case maximum
= 0.10606 mm
Largest extrusion gap, diametral for a radial seal and axial for a face seal
[77] gap_{min} = smallest gap over the temperatures
= mean 0.068 mm, worst-case minimum
= 0.029984 mm
Smallest extrusion gap
[79] gap_{allow} = permitted extrusion gap
= entered 0.2 mm
= 0.2 mm
Permitted extrusion gap, diametral for a radial seal and axial for a face seal (your record or value)

[80] u_{gap} = gap_{max} / gap_{allow}
= 0.10606 / 0.2
= 0.53032
Largest gap against the permitted gap
[81] $p_{sq,min}$ = fraction of 20000 samples with squeeze below the lower limit (cold end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: squeeze below the lower limit (cold end)
[82] $cpk_{sq,min}$ = limit distance over three σ of the linearised model (the Monte Carlo itself ru
model)
= σ = 0.0080809
= 1.5612
Capability index C_{pk}: squeeze below the lower limit (cold end)
[83] $p_{sq,max}$ = fraction of 20000 samples with squeeze above the upper limit (hot end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: squeeze above the upper limit (hot end)
[84] $cpk_{sq,max}$ = limit distance over three σ of the linearised model (the Monte Carlo itself ru
model)
= σ = 0.0077697
= 3.4698
Capability index C_{pk}: squeeze above the upper limit (hot end)
[85] $p_{fill,max}$ = fraction of 20000 samples with gland fill above the upper limit (hot end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: gland fill above the upper limit (hot end)
[86] $cpk_{fill,max}$ = limit distance over three σ of the linearised model (the Monte Carlo itself
model)
= σ = 0.01515
= 1.648
Capability index C_{pk}: gland fill above the upper limit (hot end)
[87] $p_{fill,min}$ = fraction of 20000 samples with gland fill below the lower limit (cold end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: gland fill below the lower limit (cold end)
[88] $cpk_{fill,min}$ = limit distance over three σ of the linearised model (the Monte Carlo itself
model)
= σ = 0.01376
= 2.3815
Capability index C_{pk}: gland fill below the lower limit (cold end)
[89] $p_{stretch}$ = fraction of 20000 samples with installed stretch outside its limits
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: installed stretch outside its limits
[90] $cpk_{stretch}$ = limit distance over three σ of the linearised model (the Monte Carlo itself m
model)
= σ = 0.0027575
= 3.158
Capability index C_{pk}: installed stretch outside its limits
[91] p_{gap} = fraction of 20000 samples with extrusion gap above the permitted gap
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: extrusion gap above the permitted gap

[92] cpk_{gap} = limit distance over three σ of the linearised model (the Monte Carlo itself runs model)
= σ = 0.0091585
= 4.8028
Capability index C_{pk}: extrusion gap above the permitted gap
[93] pareto = largest contributor to the squeeze spread (worst case)
= ring cross-section W 66.373 %; ring inside diameter d 9.3264 %; bore D_s 9.6019 %; piston gr D_g 14.699 %
= "ring cross-section W"
Contribution Pareto

5. Results

SYMBOL	VALUE	DESCRIPTION
$squeeze_{min}$	15.13 %	Smallest squeeze
$squeeze_{max}$	25.22 %	Largest squeeze
$fill_{max}$	83.78 %	Largest gland fill
stretch	3.510 %	Largest installed stretch
$squeeze_{nom}$	19.54 %	Squeeze at the mean dimensions
$fill_{min}$	63.95 %	Smallest gland fill
$stretch_{nom}$	2.612 %	Installed stretch, mean dimensions
$stretch_{min}$	1.729 %	Smallest installed stretch
X_c	4.158 %	Largest stretch at the cold end
red_{nom}	1.281 %	Cross-section reduction at the mean stretch
gap_{min}	0.02998 mm	Smallest extrusion gap
gap_{max}	0.1061 mm	Largest extrusion gap, diametral for a radial seal (over the temperature range, with the bore dilation)
gap_{allow}	0.2000 mm	Permitted extrusion gap, diametral for a radial seal
$p_{sq,min}$	0 %	Fraction outside the limit: squeeze below the lower limit (cold end)
$cpk_{sq,min}$	1.5612	C_{pk}: squeeze below the lower limit (cold end)
$p_{sq,max}$	0 %	Fraction outside the limit: squeeze above the upper limit (hot end)
$cpk_{sq,max}$	3.4698	C_{pk}: squeeze above the upper limit (hot end)
$p_{fill,max}$	0 %	Fraction outside the limit: gland fill above the upper limit (hot end)
$cpk_{fill,max}$	1.648	C_{pk}: gland fill above the upper limit (hot end)
$p_{fill,min}$	0 %	Fraction outside the limit: gland fill below the lower limit (cold end)
$cpk_{fill,min}$	2.3815	C_{pk}: gland fill below the lower limit (cold end)
$p_{stretch}$	0 %	Fraction outside the limit: installed stretch outside its limits
$cpk_{stretch}$	3.158	C_{pk}: installed stretch outside its limits

SYMBOL	VALUE	DESCRIPTION
p_{gap}	0 %	Fraction outside the limit: extrusion gap above the permitted gap
cpk_{gap}	4.8028	C_{pk}: extrusion gap above the permitted gap
$u_{stretch}$	0.59	Largest stretch against its limit
u_{gap}	0.53	Largest gap against the permitted gap

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Squeeze minimum within the band	$squeeze_{min}$ 15.125 %	≥ 15 %	–	[5](record)
PASS	Squeeze maximum within the band	$squeeze_{max}$ 25.225 %	≤ 30 %	–	[5](record)
PASS	Gland fill maximum within the band	$fill_{max}$ 83.783 %	≤ 85 %	–	[5](record)
PASS	Gland fill minimum within the band	$fill_{min}$ 63.953 %	≥ 60 %	–	[5](record)
PASS	Installed stretch not above the limit	stretch 3.5104 %	≤ 6 %	–	[6](rule)
PASS	Installed stretch not above the limit at the cold end	X_c 4.1575 %	≤ 6 %	–	[5](record)
PASS	Installed stretch not below the limit	$stretch_{min}$ 1.729 %	≥ 0 %	–	[5](record)
PASS	Extrusion gap within the permitted gap	gap_{max} 0.10606 mm	≤ 0.2 mm (entered 0.2 mm)	–	[7](record)

7. Applicability

WARNING records: tabulated values not taken from a reference record are α (Material properties); d_r, $d_{r,tol}$, W, W_{tol} (Ring size); $q_{lo,s}$, $q_{hi,s}$, $fill_{lo}$, $fill_{hi}$, x_{lo}, x_{hi} (Squeeze, fill and stretch bands); gap_{allow} (Permitted extrusion gap). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
$face_p$: not entered, so the default of 'External (ring ID seated on the groove)' is used.
h_f: not entered, so the default of 2.7 mm is used.
$h_{f,u}$: not entered, so the default of 0.05 mm is used.
$h_{f,l}$: not entered, so the default of 0 mm is used.
e_f: not entered, so the default of 0.05 mm is used.
$e_{f,u}$: not entered, so the default of 0.05 mm is used.
$e_{f,l}$: not entered, so the default of 0 mm is used.
ecc: not entered, so the default of 0 mm is used.
v: not entered, so the default of 0.49 is used.
$q_{lo,d}$: not entered, so the default of 10 % is used.
$q_{hi,d}$: not entered, so the default of 25 % is used.

od_{hi}: not entered, so the default of 6 % is used.
dD_p: not entered, so the default of 0 mm is used.

8. Verification

```
Template fp.oring-gland@1.4.0 Content hash f2f0e8460feb08f6 Engine calc-engine/1.0.0
Known-answer cases passed 13/13 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Parker O-Ring Handbook ORD 5700.
2. Parker PTD 5705.
3. ISO 3601-1.
4. ISO 3601-2.
5. oring.criteria.
6. oring.criteria record; PTD 5705 installation.
7. oring.extrusion.